

Arsenal FC • Revenue Uplift Calculation

How We Arrive at £4.53M Annual Uplift at 10% Dynamic Inventory

PwC × Arsenal • For Partner Review • 2025

PART 1 — THE CALCULATION WITH ASSUMPTIONS

How £4,534,051 is derived, step by step

STEP 1 — INVENTORY DECISION (Assumption)

Assumption: 10% of Emirates capacity is released for dynamic pricing per match.

Total capacity:	60,704 seats
Dynamic inventory:	10% = 6,070 seats per match
Zone split (of 6,070):	
Upper Centre	60% upper tier × 60% = 2,185 seats
Upper Corner	60% upper tier × 40% = 1,456 seats
Lower Centre	40% lower tier × 50% = 1,214 seats
Lower Wing/Cor	40% lower tier × 50% = 1,214 seats

What replaces this in production: The dynamic inventory % is a commercial decision made by Arsenal. The zone split would be calibrated to actual seat-section capacities from the stadium map.

STEP 2 — STAKES SCORE (Computed from simulated league table)

For each of the 19 home fixtures, the stakes engine reads the league table at that matchweek and scores the match on four dimensions — title race, top-four battle, European spots, relegation threat — weighted by urgency (how deep into the season we are). The maximum of the four scores becomes the stakes score (0-100) and maps to a classification label.

Assumption: The league table is simulated via a seeded random season model. In production this would be replaced by the live Premier League standings from an API feed.

STEP 3 — FILL RATE (Assumption-based formula)

- Assumption:** Base fill rates by category:

 - Cat A (big-six): 99.0%
 - Cat B (established mid): 95.0%
 - Cat C (smaller clubs): 85.0%

Adjusted upward by: $(\text{home form} - 5) \times 0.005 + (\text{star power} - 5) \times 0.004 + (\text{stakes score} / 100) \times 0.05$
Adjusted downward by: $\max(0, (\text{away distance} - 200) / 800) \times 0.02$

Default values used: home_form=7, star_power=6, distance=250

What replaces this in production: The two-layer forecasting engine (STL + SARIMA structural baseline, LightGBM feature-adjusted layer) outputs a fill rate per zone per fixture, trained on 3-4 seasons of Arsenal historical gate data.

STEP 4 — WTP AGGRESSIVENESS MULTIPLIER (Assumption)

Assumption: Base WTP multiplier by category:

- Cat A: 1.4×
- Cat B: 1.2×
- Cat C: 1.1×

Adjusted upward by: $(\text{stakes score} / 100) \times 0.5 + 0.05$ (star power bonus)

Hard bounds: min 1.0×, max 2.5×

This gives a unique WTP multiplier per fixture ranging from 1.39× (Bournemouth, Standard) to 1.86× (Man City, Title Decider).

What replaces this in production: The conjoint survey engine (HB-MNL) directly outputs WTP bounds per zone per opponent tier. The multiplier becomes an observed quantity, not an assumption.

STEP 5 — PRICE UPLIFT CEILING (Assumption)

Assumption: Maximum £ uplift per seat above base price, by category × stakes classification:

Category	Title Decider	High Stakes	Standard
Cat A	£75	£55	£15
Cat B	£38	£28	£8
Cat C	£22	£13	£4

Derby/rivalry bonus on top:

- NLD (Tottenham): +£20
- Big-six (City, United, Liverpool, Chelsea): +£10

Effective ceiling = category ceiling × WTP multiplier Example: Man City = $(£75 + £10) \times 1.86 = £158.10$ effective max uplift per seat (averaged across zones)

The ceiling scales proportionally with each zone's base price — Upper Centre absorbs more absolute £ than Lower Wing.

What replaces this in production: Conjoint-derived WTP ceiling per zone. The £75 / £55 / £15 values are starting assumptions that the HB-MNL survey will replace with measured numbers.

STEP 6 — OVERSUBSCRIPTION REGIME (Key logic)

When fill rate $\geq 95\%$, Arsenal cannot add more fans regardless of price. In this regime, the LP suspends price-elasticity and floats price to the WTP ceiling — quantity is held fixed, revenue gain is purely from the price increase.

Effect: 14 out of 19 fixtures operate in oversubscription regime. This is realistic — Emirates averages 99.3% utilisation. In this regime, every £1 of price increase on a dynamic seat is pure additional revenue with zero attendance risk.

The 5 Cat C fixtures (Bournemouth, Southampton, Everton, Ipswich, Leicester) operate in the elastic regime — fill rate below 95%, so the LP balances price increase against attendance risk using the zone elasticity coefficients.

STEP 7 — LP OPTIMISER (30-point price grid per zone)

For each zone per fixture, the optimiser searches a 30-point price grid from base price to base price + zone-specific ceiling. It selects the price that maximises zone revenue subject to the fill rate and ceiling constraints.

Zone elasticity coefficients (assumption from conjoint defaults):

- Upper Centre: -0.45 (most inelastic — premium panoramic views, wealthier demographic)
 - Upper Corner: -0.55
 - Lower Centre: -0.75
 - Lower Wing/Corner: -0.85 (most elastic — cheapest seats, most price-sensitive)
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STEP 8 — PER-MATCH RESULTS

MW	Opponent	Cat	Stakes	Score	Fill Rate	WTP	Uplift
3	Newcastle	A	Standard	14.6	99.9%	1.52×	£128,071
9	Aston Villa	B	Standard	27.5	97.7%	1.39×	£57,704
11	Fulham	B	European Spot	30.3	97.8%	1.40×	£58,203
17	West Ham	B	European Spot	39.1	98.2%	1.45×	£60,553
21	Man Utd	A	High Stakes	56.5	99.9%	1.73×	£631,646
22	Bournemouth	C	European Spot	48.9	88.7%	1.39×	£9,403
23	Southampton	C	High Stakes	50.8	88.8%	1.40×	£20,685
24	Everton	C	High Stakes	52.8	88.9%	1.41×	£20,764
25	Ipswich	C	High Stakes	54.7	89.0%	1.42×	£20,841
26	Brentford	B	High Stakes	56.6	99.1%	1.53×	£225,620
27	Nottm Forest	C	High Stakes	58.6	89.2%	1.44×	£20,991
28	Tottenham	A	High Stakes	67.8	99.9%	1.79×	£754,100
29	Wolves	B	High Stakes	62.5	99.4%	1.56×	£230,729
32	Man City	A	Title Decider	81.4	99.9%	1.86×	£888,068
33	Crystal Palace	B	High Stakes	70.3	99.8%	1.60×	£237,573
34	Chelsea	A	High Stakes	72.2	99.9%	1.81×	£660,855
35	Brighton	B	High Stakes	74.2	99.9%	1.62×	£240,808
36	Liverpool	A	Top Four Decider	76.1	99.9%	1.83×	£256,984
38	Leicester	C	Top Four Decider	80.0	90.3%	1.55×	£10,453

SEASON TOTAL

Total annual uplift:	£4,534,051
Dynamic seats per match:	6,070
Total dynamic seat-matches (×19):	115,330
Average uplift per seat per match:	£39.31

Where the uplift comes from:

- Top 4 fixtures (Man City, Tottenham, Chelsea, Man Utd) = **£2.93M — 65% of total**
- Mid-tier Cat B fixtures (Brentford, Wolves, Crystal Palace, Brighton) = **£0.93M — 20%**
- Cat A early season (Newcastle) + Liverpool = **£0.39M — 9%**
- All 6 Cat C fixtures combined = **£0.10M — 2%**

Key sensitivity: Cat C fixtures barely contribute because their uplift ceilings are low and fill rates below 95% mean the elastic regime limits price increases. If the conjoint survey reveals higher WTP for Cat C than assumed, this number moves meaningfully.

PART 2 — ANTIGRAVITY PROMPT

Connect this calculation to the real engines

Paste this into Antigravity:

I have a working revenue uplift calculation for Arsenal FC's dynamic pricing system that produces £4,534,051 annual uplift at 10% dynamic inventory across 19 home fixtures. This is built on a set of assumptions that need to be replaced by the actual production engines already coded in the project. I need you to trace through the calculation and show exactly how each assumption gets replaced.

The calculation has five assumption-driven inputs:

1. Fill rate per fixture — currently a formula: base fill rate (Cat A=99%, B=95%, C=85%) adjusted by hardcoded feature coefficients ($\text{home_form} \times 0.005$, $\text{star_power} \times 0.004$, $\text{stakes_score}/100 \times 0.05$). This needs to be replaced by the output of `forecasting_engine_epl.py` — specifically the `EPLForecastingEngine` class which produces `overall_fill_rate` per fixture from a two-layer model (STL + SARIMA structural layer, LightGBM feature-adjusted layer). Show what input record the engine expects (referencing `_prepare_layer2_data` and the feature set it builds) and what `overall_fill_rate` value it would produce for each of the 19 fixtures. Use the same 19 fixtures and matchweeks from the current schedule.

2. WTP aggressiveness multiplier — currently: Cat A base 1.4×, Cat B 1.2×, Cat C 1.1×, adjusted by $(\text{stakes_score}/100) \times 0.5$. This needs to be replaced by the conjoint engine

output. The conjoint engine (`conjoint_engine_ep1.py` if it exists, or describe what it should produce) should output a WTP multiplier per zone per opponent tier derived from the HB-MNL conjoint survey. Show what the conjoint engine's output record looks like and how the WTP multiplier maps from that record into the LP.

3. Price uplift ceiling — currently hardcoded: Cat A Title Decider £75, High Stakes £55, Low £15; Cat B Title Decider £38, High Stakes £28, Low £8; Cat C Title Decider £22, High Stakes £13, Low £4. These need to be replaced by `zone_price_bounds` from the conjoint engine's WTP data (`ep1_wtp_big_six.json`, `ep1_wtp_mid.json`, `ep1_wtp_small.json`). Show how `EPLLOptimizer._load_json()` reads these files and how `bounds.get(c_zone)` maps to floor and ceiling per zone in `lp_optimizer_ep1.py`.

4. Zone elasticity coefficients — currently hardcoded: Upper Centre -0.45, Upper Corner -0.55, Lower Centre -0.75, Lower Wing -0.85. These come from the conjoint survey. Show where in `lp_optimizer_ep1.py` these are applied (the `elasticities` dict in `optimize()`) and confirm they match or differ from the demo assumptions.

5. Stakes score — currently computed inline in the demo from a simulated league table. The real engine is `StakesEngine` in `stakes_engine.py` with `compute_stakes_score()`. Show how this method takes `home_position`, `away_position`, `home_points`, `away_points`, and the full `standings` list as inputs, and how its output `match_stakes_score` and `match_stakes_label` feed into the LP as the demand signal.

For each of the five inputs, produce:

- What the assumption currently is in the demo calculation
- What the real engine produces instead
- What input data that engine needs to run (what fields, from what source)
- The exact line or method in the production code that replaces the assumption
- A concrete example value for the Man City MW32 fixture specifically, since that is the single highest-value fixture at £888,068

Then rerun the full 19-fixture uplift calculation substituting the real engine outputs where available and assumptions where the engine requires data we don't yet have (e.g. the conjoint survey hasn't been run yet). Show the revised per-match table and revised annual total, clearly marking which inputs are now engine-derived vs still assumed. The goal is to show a PwC media partner exactly how the £4.53M number will change — and in which direction — when the real engines replace the demo assumptions.